

Emmanouil Karantoumanis

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📍 Kozani, Greece | 📅 20/05/1996

EDUCATION

- Postdoctoral Researcher in Electrical and Computer Engineering** 2026 – Present
at the Department of Electrical and Computer Engineering, University of Western Macedonia Kozani, Greece
- Research focus: **Optimization, Mathematical Programming, Algorithms, Machine Learning, and Artificial Intelligence**
- Ph.D. in Electrical and Computer Engineering** 2019 – 2026
at the Department of Electrical and Computer Engineering, University of Western Macedonia Kozani, Greece
- Thesis Title: **Optimization algorithms for black-box problems**
 - Research Interests: Mathematical Programming, Machine Learning, Artificial Intelligence
- B.Sc. in Electrical and Computer Engineering** 2014 – 2019
at the Department of Electrical and Computer Engineering, University of Western Macedonia Kozani, Greece
- Grade: **7.04/10**
 - Diploma Thesis: **Energy consumption estimation in cloud computing** (Grade: 10/10)

EXPERIENCE

- University of Western Macedonia** May 2024 – Present
Researcher - Project: "SYNERGINN EDIH" Kozani, Greece
- Designed and implemented AI digital services for SMEs in the energy and environment sectors.
 - Developed specialized software applications and prepared educational material for digital innovation seminars.
 - Designed and implemented the project website, ensuring clear communication of objectives, services, and results.
 - Developed and configured the projects Moodle e-learning platform, structuring digital courses and training content for SMEs.
- University of Western Macedonia** Nov. 2023 – Sept. 2025
Researcher - Project: "Interconnected Smart Cities for Greece 2.0" Kozani, Greece
- Design and implementation of a web-based integration platform for vertical smart city solutions.
 - Developed an application for optimal waste collection routing using Machine Learning techniques.
- Chamber of Commerce and Industry of Kozani** Nov. 2023 – Oct 2025
External Collaborator - Project: "TOTEM" (Erasmus+) Kozani, Greece
- Implemented and configured the Moodle e-learning platform to support the project's educational objectives.
- University of Western Macedonia** June 2024 – May 2025
Researcher - Project: "ANALYST: A New Advanced Level for Your Specialised Training" Kozani, Greece
- Developed and customized the Moodle e-learning platform for cybersecurity and data analytics training.
- University of Western Macedonia** July 2021 – Sept. 2023
Researcher - Project: "AGROTOUR: Regional Excellence in Western Macedonia" Kozani, Greece
- Contributed to the development of new technologies for Agri-food and Tourism sectors under the NSRF 2014-2020 framework.
 - Developed and implemented object detection algorithms for the early prediction of plant diseases, supporting smart agriculture applications.

- Designed and implemented the network infrastructure and pilot digital applications for the Municipality's smart city initiative.

TEACHING EXPERIENCE

Teaching Assistant & Lecturer

- Lecturer at MSc "Digital Games Development": Course **Algorithmic Thinking** (2024-2025).
- Lecturer at MSc "Digital Games Development": Course **Algorithmic Thinking** (2025-2026).
- Teaching Assistant for undergraduate courses: "**Analysis and Design of Algorithms**", "**Discrete Mathematics**", and "**Algorithms and Data Structures**"(2020-2026).

SKILLS

Programming Languages: Python, C, C++, MATLAB, SQL, PHP, JAVASCRIPT, HTML, CSS

Languages: English (Proficiency - C2), German (Intermediate - B1), Greek (Native)

IMPLEMENTED MOODLE PLATFORMS

- <https://elearning-analystproject.eu>
- <http://elearning.synerginn.eu>
- <https://totem.newcotech.io>
- <https://mychoice.app.synerginn.eu>

PUBLICATIONS

- Karantoumanis, E., Zygidis, T., & Ploskas, N. (2025). A Surrogate-Based Adaptive Sampling Approach for Electromagnetic Problems. **International Journal of Numerical Modelling: Electronic Networks, Devices and Fields**, 38(5), e70121. [cite: 108, 109]
- Karantoumanis, E., & Ploskas, N. (2025). A surrogate-based adaptive sampling approach for mixed-integer black-box optimization problems. **Journal of Optimization Theory and Applications (JOGO)**. [cite: 107]
- Karantoumanis, E., & Ploskas N. (2024). Optimizing concentrated solar power plants using adaptive sampling-based derivative-free algorithms. **In 28th Pan-Hellenic Conference on Informatics (PCI 2024)**. [cite: 105, 106]
- Karantoumanis, E., Balafas, V., Louta, M., & Ploskas, N. (2024). Real-time plant disease detection from a small image dataset using data augmentation and deep learning methods. **In Soft Computing (SOCO)**. [cite: 104]
- Karantoumanis, E., & Ploskas, N. (2024). Improving derivative-free optimization algorithms through an adaptive sampling procedure. **Results in Control and Optimization**, 100460. [cite: 102, 103]
- Balafas, V., Karantoumanis, E., Louta, M., & Ploskas, N. (2023). Machine learning and deep learning for plant disease classification and detection. **In IEEE Access**, doi: 10.1109/ACCESS.2023.3324722. [cite: 101]
- Karantoumanis, E., Balafas, V., Louta, M., & Ploskas, N. (2022). Computational comparison of image preprocessing techniques for plant diseases detection. **In 7th SEEDA-CECNSM (pp. 1-5). IEEE**. [cite: 95, 96, 97]
- Louta, M., et al. (2022). Intelligent Pesticide and Irrigation Management in Precision Agriculture: The case of VELOS Project. **In HAICTA 2022, Athens, Greece**. [cite: 91, 92, 93, 94]
- Karantoumanis, E., Krallis, X., & Ploskas N. (2021). An experimental evaluation of sensors on a LoRaWAN-based GPS tracking system. **In 25th Pan-Hellenic Conference on Informatics (PCI 2021). ACM**, 224228. [cite: 88, 89, 90]

- Karantoumanis, E., & Ploskas, N. (2020). Power consumption predictors for software running in data centers. **Annals of Mathematics and Artificial Intelligence (under review)**. [cite: 87, 88]
- Karantoumanis, E., & Ploskas, N. (2020). Power consumption estimation in data centers using machine learning techniques. **LION 2020. Lecture Notes in Computer Science, vol 12096. Springer, Cham**. [cite: 84, 85, 86]